

HKUST MATH 2431 - problem set 6

3.

3. (*) Hazard rate

[4 Points]

We model the life-time of a device by a continuously distributed random variable $T \geq 0$ with cumulative distribution function $F(t)$ and probability density function $f(t)$. We assume that f is continuous in every $t \geq 0$. The *hazard rate* is defined as

$$\lambda(t) = \lim_{h \searrow 0} \frac{\mathbf{P}[t \leq T \leq t+h | T \geq t]}{h}, \quad t \geq 0$$

(this is also relevant for life-insurance purposes).

(a) Express $\lambda(t)$ in terms of $f(t)$ and $F(t)$.

3. a. $\lambda(t) = \lim_{h \searrow 0} \frac{\mathbf{P}[t \leq T \leq t+h | T \geq t]}{h}$

$$= \lim_{h \searrow 0} \frac{1}{h} \frac{\mathbf{P}[t \leq T \leq t+h]}{\mathbf{P}[T \geq t]}$$
$$= \lim_{h \searrow 0} \frac{1}{h} \frac{F(t+h) - F(t)}{1 - F(t)}$$
$$= \frac{f(t^+)}{1 - F(t)} = \frac{f(t)}{1 - F(t)}$$

def of derivative from the right

note: $\{\omega \in \Omega : t \leq T \leq t+h\} \subseteq \{\omega \in \Omega : T \geq t\}$

$\mathbf{P}[T \leq x] = \mathbf{P}[T < x] \Leftrightarrow F(t) \text{ is abs. cont.} \Leftrightarrow f(t) \text{ is cont.}$

$f \text{ is cont.} \Rightarrow f(t^+) = f(t)$

(b) Suppose that $T \sim \mathcal{E}(\alpha)$ with $\alpha > 0$, and calculate $\lambda(t)$.

$$3.b. T \sim \exp(\alpha) \Rightarrow f(t) = \alpha e^{-\alpha t} \mathbb{1}_{\text{tag}(t)}$$

$$(t \geq 0) \quad F(t) = (1 - e^{-\alpha t}) \mathbb{1}_{[0, +\infty)}(t)$$

$$\lambda(t) = \frac{f(t)}{1 - F(t)} = \frac{\alpha e^{-\alpha t}}{e^{-\alpha t}} = \alpha$$

(c) Verify that $F(t) = 1 - \exp\left(-\int_0^t \lambda(s) ds\right)$, when $t \geq 0$.

3. c.

$t < 0$: both sides = 0

$$t \geq 0 : 1 - \exp\left(-\int_0^t \lambda(s) ds\right)$$

$$= 1 - \exp\left(-\int_0^t \frac{f(s)}{1-F(s)} ds\right)$$

$$= 1 - \exp\left(-\int_{s=0}^{s=t} \frac{1}{1-F(s)} dF(s)\right)$$

$$= 1 - \exp\left(\left[\ln(1-F(s))\right]_0^t\right)$$

$$= 1 - \exp\left(\ln\left(\frac{1-F(t)}{1-F(0)}\right)\right)$$

$$= 1 - \exp(\ln(1-F(t)))$$

$$= 1 - (1-F(t))$$

$$= F(t)$$

$$F(0) = 0$$

$f = f(t)$ has
support on $t \geq 0$

(d) Calculate $F(t)$ and $f(t)$ for $t \geq 0$ when $\lambda(t) = \alpha t^\gamma$ ($\alpha > 0, \gamma > 0$) (Weibull distribution).

$$3.d. \quad F(t) = 1 - \exp\left(-\int_0^t \lambda(s) ds\right) \quad (\text{assume } t \geq 0)$$

$$\int_0^t \lambda(s) ds = \int_0^t \alpha s^\gamma ds$$

$$= \frac{\alpha}{\gamma+1} \left[s^{\gamma+1} \right]_0^t$$

$$= \frac{\alpha t^{\gamma+1}}{\gamma+1}$$

$$F(t) = 1 - \exp\left(-\frac{\alpha t^{\gamma+1}}{\gamma+1}\right)$$

$$f(t) = F'(t) = \alpha t^\gamma \exp\left(-\frac{\alpha t^{\gamma+1}}{\gamma+1}\right)$$

(e) For $T \geq 0$, show that $\mathbf{E}[T] = \int_0^\infty \exp\left(-\int_0^t \lambda(s) ds\right) dt$.

3.e. assume $T \geq 0$

$$E[T] = \int_0^{+\infty} t f(t) dt \quad (\text{support on } [0, +\infty))$$

$$= \lim_{s \rightarrow +\infty} \left([t F(t)]_0^s - \int_0^s t f(t) dt \right)$$

$$= \lim_{s \rightarrow +\infty} \left(s F(s) - \int_0^s \left(1 - \exp\left(-\int_0^t \lambda(u) du\right) \right) dt \right)$$

$$= \lim_{s \rightarrow +\infty} \left(\underbrace{s F(s)}_{\text{Assume}} - s + \int_0^s \exp\left(-\int_0^t \lambda(u) du\right) dt \right)$$

$$\rightarrow s(F(s)-1) : \text{Assume } s \rightarrow +\infty, s(F(s)-1) \rightarrow 0$$

$$= \int_0^{+\infty} \exp\left(-\int_0^t \lambda(s) ds\right) dt$$

extra note!

$$\lim_{s \rightarrow +\infty} s(F(s)-1)$$

$$= \lim_{s \rightarrow +\infty} s \underbrace{\left(-\exp\left(-\int_0^s \lambda(t) dt\right) \right)}_{\rightarrow 0}$$

$$= \lim_{s \rightarrow +\infty} - \frac{s}{\exp\left(\int_0^s \lambda(t) dt\right)} \quad s \rightarrow +\infty$$

$$= \lim_{s \rightarrow +\infty} - \frac{1}{\lambda(s) \exp\left(\int_0^s \lambda(t) dt\right)}$$

Ideally, the limit should be zero. However, this need not be the case. Consider $\lambda(t) = \begin{cases} 1, & t \in [0, 1] \\ \frac{1}{t}, & t > 1 \end{cases}$

$\lambda(t) \geq 0$ and $\int_0^\infty \lambda(t) dt = 1 + [\ln(t)]_1^\infty = +\infty$
 when $t \geq 0$
 (so that $f(t)$ is monotonic) (so that as $t \rightarrow +\infty$, $f(t) \rightarrow 1$)
 But the limit above has

$$= \lim_{s \rightarrow +\infty} - \frac{s}{\exp(1 + \ln(s))} = \lim_{s \rightarrow +\infty} - \frac{1}{e} \frac{s}{s} = -\frac{1}{e}$$

So for the result to hold, we additionally
 require $\lambda(t)$ to not decay too quickly.
 alternatively
 (or, the result always hold if we treat
 both sides $\rightarrow +\infty$ as "equal")

4.

4. (*) More on expectation

[4 points]

- (a) Let $\mathcal{S}_n$ be set of all permutations of $\{1, \dots, n\}$ and $\mathbf{P} = \mathcal{U}(\mathcal{S}_n)$ be the uniform distribution on $\mathcal{S}_n$. For every permutation $\omega \in \mathcal{S}_n$, let $X(\omega)$ be the number of fixed points of ω . Find $\mathbf{E}[X]$.
 Hint: Consider for $i = 1, \dots, n$, $A_i = \{\omega \in \mathcal{S}_n : \omega(i) = i\}$. Then $X = \sum_{1 \leq i \leq n} \mathbf{1}_{A_i}$.

4.a Define A_i as in the hint

$$|A_i| = (n-1)!$$

$$E[\mathbb{1}_{A_i}] = \frac{(n-1)!}{n!} = \frac{1}{n}$$

$$E[x] = E\left[\sum_{i=1}^n \mathbb{1}_{A_i}\right] = \sum_{i=1}^n E[\mathbb{1}_{A_i}] = 1$$

- (b) In a fictional company, a worker's legal code specifies as a holiday any day during which at least one worker has their birthday. All other days are working days. How many workers must the factory employ so that the expected number of working worker-days is maximized during the year? (Worker-days = Headcount $\times$ Days.)

Hint: First consider the expectation of the total number of birthday-free days.

4.b. Assume 365 days, labeled $\{1, \dots, 365\}$.

Define $\mathbb{I}_{d,n}$ as whether the worker n have birthday on day d .

Define $\mathbb{I}_d$ as whether day d is working day.

$$\mathbb{I}_d = \prod_{1 \leq i \leq n} (1 - \mathbb{I}_{d,i})$$

notice $\mathbb{I}_{d,i}$ are mutually independent among $i \in \{1, \dots, n\}$ for fixed d .

$$E[\mathbb{I}_d] = \left(1 - \frac{1}{365}\right)^n = \left(\frac{364}{365}\right)^n$$

So number of working days = $365 \left(\frac{364}{365}\right)^n$ (linearity of expectation, NOT require independence)

$$\text{expected worker-days} = 365 n \left(\frac{364}{365}\right)^n$$

$$\Delta(n) = 365(n+1)\left(\frac{364}{365}\right)^{n+1} - 365n\left(\frac{364}{365}\right)^n$$

$$= \left(\frac{364}{365}\right)^n 365 \left(\frac{364(n+1)}{365} - n\right)$$

$\Delta(n)$ is forward difference

$$= \frac{364}{365} (n+1) - n$$

$$= \frac{364}{365} - \frac{n}{365}$$

$$\begin{aligned} &> 0, n \leq 363 \\ \text{So } \Delta(n) &= 0, n = 364 \\ &< 0, n \geq 365 \end{aligned}$$

So $n=364$ or $n=365$ maximizes expected worker days.